Algebraic extinction in acyclic compartment models

A sharp four-state counterexample, slow-tail families and a spectral criterion

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Abstract

Chan, Crommelin and Mandjes conjectured that vanishing states in a conservative quadratic interaction model decay exponentially when its transition graph is acyclic and each dependency cycle contains a vertex with a persistent dependency successor. We disprove this conjecture with a four-state model having positive integer rates and strictly positive rational initial fractions. Its vanishing state satisfies $I(t) \sim 18t^{-2}$ and $\int_0^\infty I(t)\,dt = \log 2$. We prove that four states are necessary. For every integer $m \geq 2$, we construct an $(m+2)$-state model satisfying both graph assumptions with $I(t) \sim K_m t^{-m/(m-1)}$. The exponents approach one, so the assumptions give no dimension-independent power improvement over the integrability threshold. We classify the four-state model's full extinction boundary and prove a trajectory-dependent repair: under transition acyclicity, all vanishing coordinates decay exponentially if and only if their limiting transverse Jacobian is Hurwitz. The aggregate vanishing population has logarithmic decay rate equal to that matrix's spectral bound. Exact rational checks accompany the analytic proofs. The constructions are existence results; they do not establish that algebraic extinction is generic or describe an observed epidemic.

1. The question and its mathematical scope

When a compartment empties, convergence to zero does not by itself say how long its remaining activity matters. An exponential tail has a fixed proportional decay rate. An algebraic tail slows continually, and can take much longer to cross a small stopping threshold even when its total future activity is finite. This paper asks whether two natural graph conditions distinguish these possibilities in a conservative interaction model. The answer is no: a positive removal mechanism can be cancelled, at the limiting state, by positive inflow.

The argument has four stages. Section 2 gives a fully explicit counterexample that can be checked without the later theory. Sections 3 and 4 establish how slow the tail can be and why fewer than four compartments cannot produce the obstruction. Section 5 explains the example's sensitivity to initial data. Section 6 then identifies the missing trajectory-level condition and its exact logarithmic rate. The graph conditions concern admissible contacts and transitions; the spectral condition additionally uses the limiting population fractions.

Chan, Crommelin and Mandjes [1, Sections 2.2 and 3.3] study interaction-driven population models and formulate an exponential-decay conjecture under two graph conditions. Their Theorem

2 establishes integrability of vanishing states under those conditions. The distinction between integrability and exponential decay is the issue addressed here.

We use a conversion-channel notation. A channel

$$i \longrightarrow j \quad [k; a], \qquad i \neq j,\ a > 0, \tag{1}$$

moves an individual from state i to state j, after contact with state k, at flux $a y_i y_k$. The contact partner remains in state k. Each channel subtracts this flux from $\dot{y}_i$ and adds it to $\dot{y}_j$. The state fractions start strictly positive and sum to one. A finite set of channels defines a conservative quadratic ordinary differential equation

$$\dot{y} = F(y). \tag{2}$$

The source convention is recovered by setting $\beta^j_{ik} = a$ for a destination $j \neq i$, adding coefficients when channels coincide, and setting $\beta^i_{ik} = \sum_{j \neq i} \beta^j_{ik}$. Nonnegative inflows and bounded outflow hazards on the invariant simplex ensure that every initially positive coordinate remains positive at every finite time.

The **transition graph** contains $i \to j$ for every channel (1). The **dependency graph** contains $i \to k$. A state is **persistent** when its trajectory tends to a strictly positive limit and **vanishing** when that limit is zero. We impose the source's conditions:

(T) The transition graph is a directed acyclic graph.

(D) Every dependency cycle contains a vertex with a persistent dependency successor.

A successor is an immediate out-neighbour. Condition (D) permits that successor to lie on the cycle. It is a condition on the model and its trajectory, because persistence is an asymptotic property. We do not strengthen it to a requirement that persistence be identifiable from graph topology alone. The conjectured conclusion is that every vanishing y_i admits constants $C_i, r_i > 0$ with $y_i(t) \leq C_i e^{-r_i t}$ for all $t \geq 0$.

Algebraic decay in epidemic models is already studied, including by Luhar et al. [2]. Polynomial kinetic realisations have a substantial literature [3], and exact time changes and first integrals are established tools [4,5]. We do not claim these methods or algebraic decay itself as new. The contribution is the explicit failure of the two specified graph assumptions to force exponential decay, the sharp minimum dimension, the family of admissible exponents, and the precise spectral repair proved below. A bounded prior-art report accompanies the manuscript.

2. An exact four-state counterexample

Consider

$$\dot{X} = -XI, \qquad \dot{Y} = -2YI, \qquad \dot{I} = (X - 2Y)I, \qquad \dot{R} = 4YI, \tag{3}$$

with

$$(X_0, Y_0, I_0, R_0) = \left(\frac{2}{9}, \frac{2}{9}, \frac{1}{18}, \frac{1}{2}\right). \tag{4}$$

The channels are $X \to I$ $[I; 1]$, $Y \to R$ $[I; 2]$, and $I \to R$ $[Y; 2]$. The two latter channels each contribute $2YI$ to $\dot{R}$; their sum explains the factor four. In the source's indices, with $(X, Y, I, R) = (1, 2, 3, 4)$, the only nonzero coefficients are

$$\beta^1_{13} = \beta^3_{13} = 1, \quad \beta^2_{23} = \beta^4_{23} = 2, \quad \beta^3_{32} = \beta^4_{32} = 2. \tag{5}$$

Theorem 1 (counterexample). The system (3)-(4) satisfies (T) and (D). Its limit is $(1/9, 1/18, 0, 5/6)$, and

$$I(t) \sim \frac{18}{t^2}, \qquad \int_0^\infty I(t)\,dt = \log 2. \tag{6}$$

Consequently Conjecture 1 of [1] is false as stated.

Proof. The sum of the derivatives in (3) is zero. The first three components have a multiplicative state factor and bounded coefficients on the simplex, so they remain positive at every finite time. The final component increases from 1/2. Define

$$\tau(t) = \int_0^t I(s)\,ds, \qquad u(t) = e^{-\tau(t)}. \tag{7}$$

Direct integration gives

$$X = \frac{2}{9}u, \quad Y = \frac{2}{9}u^2, \quad I = \frac{2}{9}\left(u - \frac{1}{2}\right)^2, \qquad \dot{u} = -\frac{2}{9}u\left(u - \frac{1}{2}\right)^2. \tag{8}$$

The formula for I follows either by differentiating it or from the conserved quantity $I + X - Y = 1/18$. The scalar solution decreases from one towards 1/2 and cannot reach that root in finite time. Writing $v = u - 1/2$ yields

$$\left(\frac{1}{v}\right)' = \frac{2}{9}\left(\frac{1}{2} + v\right) \longrightarrow \frac{1}{9}.$$

Thus $v \sim 9/t$ and $I \sim 18/t^2$. Equation (7) gives the integral in (6). Substitution into (8) and conservation give the stated limit.

The transition edges are $X \to I$, $I \to R$, and $Y \to R$, so (T) holds. The dependency edges are $X \to I$, $Y \to I$, and $I \to Y$. Its only simple cycle is $Y \to I \to Y$. The vertex I has persistent dependency successor Y, whose limit is 1/18, so (D) holds. Finally $e^{rt}I(t) \to \infty$ for every $r > 0$, which excludes every positive exponential upper bound. □

The active three-variable subsystem is a Lotka-Volterra system. Its interaction matrix is

$$\begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & -2 \\ 1 & -2 & 0 \end{pmatrix},$$

and it has the first integrals $I + X - Y$ and Y/X^2. This classical integrable structure [5] makes the example transparent; it is not the novelty claim.

3. A family of exponents approaching one

The previous obstruction is not restricted to the exponent two.

Theorem 2 (arbitrary multiplicity). For every integer $m \geq 2$, there is a conservative quadratic channel model on $m + 2$ states with positive integer rates and strictly positive rational initial fractions satisfying (T) and (D). Exactly one state, I, vanishes. Set

$$\kappa_m = \frac{1}{2\,3^m}, \quad p_m = \frac{m}{m-1}, \quad K_m = \kappa_m^{-1/(m-1)}(m-1)^{-m/(m-1)}. \tag{9}$$

Then

$$I(t) \sim K_m t^{-p_m}, \qquad \int_0^\infty I(t)\,dt = \log 2. \tag{10}$$

Construction and proof. Use states $X_1, \dots, X_m, I, R$ and let

$$P_m = \{k : m-k \text{ is odd}\}, \qquad N_m = \{k : m-k \text{ is even}\}.$$

For each $k \in P_m$, use $X_k \to I$ $[I;k]$. For each $k \in N_m$, use both $X_k \to R$ $[I;k]$ and $I \to R$ $[X_k;k]$. The equations are

$$\dot{X}_k = -kX_kI, \quad \dot{I} = I\left(\sum_{k\in P_m} kX_k - \sum_{k\in N_m} kX_k\right), \quad \dot{R} = 2I\sum_{k\in N_m} kX_k. \tag{11}$$

Take

$$X_k(0) = \kappa_m\binom{m}{k}2^k, \qquad I(0) = \kappa_m, \qquad R(0) = \frac{1}{2}. \tag{12}$$

The binomial theorem gives $\sum_k X_k(0) + I(0) = \kappa_m 3^m = 1/2$, so these are positive normalised fractions.

Let z solve

$$\dot{z} = -\kappa_m(1+z)z^m, \qquad z(0) = 1. \tag{13}$$

Separation gives

$$\kappa_m t = \int_{z(t)}^1 \frac{dv}{(1+v)v^m}. \tag{14}$$

This integral shows that z remains in $(0,1]$ at finite times and decreases to zero. The complete solution is

$$X_k = \kappa_m\binom{m}{k}(1+z)^k, \qquad I = \kappa_m z^m, \tag{15}$$

$$R = 1 - \kappa_m((2+z)^m + z^m - 1). \tag{16}$$

To check (15), differentiate and use

$$\sum_{k=1}^m (-1)^{m-k+1}k\binom{m}{k}(1+z)^k = -m(1+z)z^{m-1}. \tag{17}$$

Equation (16) follows from conservation. The initial fractions agree with (12), and polynomial ODE uniqueness identifies the parametrised trajectory with the model solution. In particular,

$$X_k^* = \kappa_m\binom{m}{k} > 0, \quad I^* = 0, \quad R^* = 1 - \kappa_m(2^m - 1) > 0. \tag{18}$$

The ordering $X_1, \dots, X_m, I, R$ is topological for the transition graph. Every reservoir has dependency edge $X_k \to I$; I has edge $I \to X_k$ exactly when $k \in N_m$. Every simple cycle is therefore $I \to X_k \to I$ for $k \in N_m$. Its vertex I has persistent successor X_k. Both conditions hold.

Finally,

$$\frac{d}{dt}z^{-(m-1)} = (m-1)\kappa_m(1+z) \longrightarrow (m-1)\kappa_m.$$

Averaging this derivative yields (10) with the constants in (9). Since

$$\frac{d}{dt}\log(1+z) = -I,$$

the time integral equals $\log 2$. □

The first instances are exact and small. Formula (12) gives the complete initial vector for each row.

States	m	Initial activity $I(0)$	Tail of $I(t)$
4	2	$1/18$	$18t^{-2}$
5	3	$1/54$	$(3\sqrt{3}/2)t^{-3/2}$
6	4	$1/162$	$2^{1/3}t^{-4/3}$

Corollary 3. For every $\delta > 0$, some model satisfying (T) and (D) has a vanishing state that is not $O(t^{-1-\delta})$.

Proof. Choose m with $1/(m-1) < \delta$ and apply Theorem 2. Then $t^{1+\delta} I(t) \to \infty$. □

This assertion ranges over dimensions. It does not exclude bounds that depend on the number of states, and it does not classify all possible non-power-law tails in the source model class.

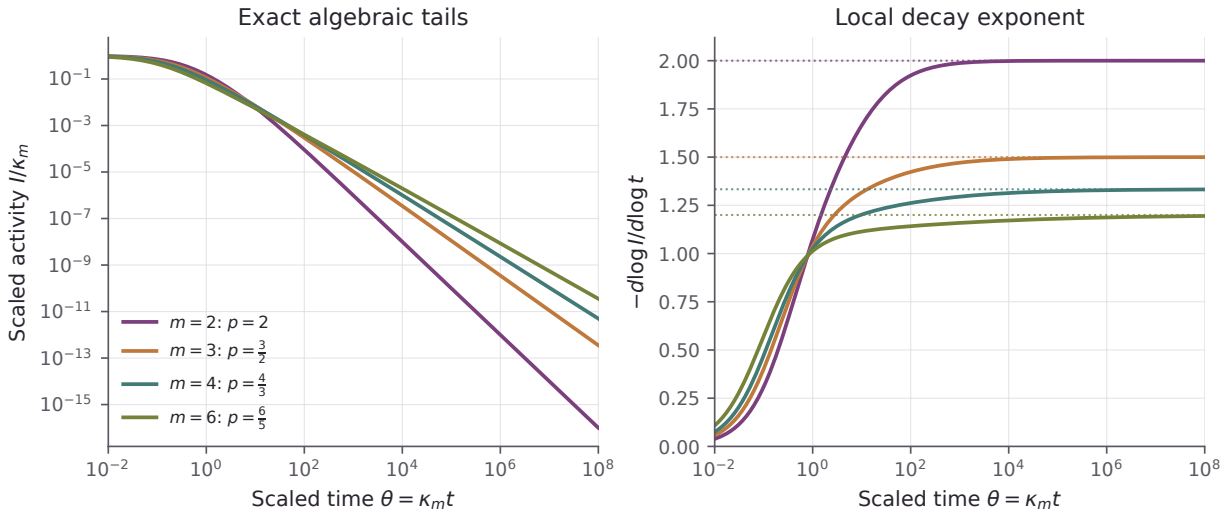


Figure 1: Numerical evaluations of the exact scalar quadrature for the constructed family. Time is rescaled as $\theta = \kappa_m t$, and activity as I/κ_m. The right panel evaluates the analytic local log-log slope; dotted levels are $m/(m-1)$.

4. Four states are necessary

We first reproduce the structural facts needed for a self-contained lower bound on dimension.

Lemma 4. Under (T), every state fraction has a limit and every channel flux has a finite integral. Under (T) and (D), every vanishing state is integrable and has a persistent dependency successor.

Proof. Topologically label the states $1, \dots, n$. Every prefix mass $\sum_{i \le j} y_i$ is nonincreasing, so all coordinate limits exist. The nonnegative potential

$$V(t) = \sum_{i=1}^{n} (n-i) y_i(t)$$

decreases by $(j-i)$ times the flux of a channel $i \to j$. Hence $-V'$ dominates the sum of all channel fluxes, which proves their integrability.

Let $\mathcal{A}$ contain the vanishing states with infinite time integral. No vertex of $\mathcal{A}$ can have a persistent dependency successor: such a successor would give outflow at least cy_i eventually, contradicting the finite flux integral. Every vertex of $\mathcal{A}$ must nevertheless have a nonintegrable dependency successor. Indeed, with outflow hazard $h_i(t) = \sum_k \beta^i_{ik} y_k(t)$, one has

$$y_i(t) \geq y_i(0) \exp\left(- \int_0^t h_i(s)\, ds \right). \tag{19}$$

If all successors were integrable, this lower bound would stay positive. A nonintegrable successor cannot be persistent by the preceding observation, so it belongs to $\mathcal{A}$. Following these edges produces a cycle with no vertex having a persistent successor, contrary to (D). Thus $\mathcal{A}$ is empty. Applying (19) again shows that every vanishing state must have a persistent successor. Persistent states cannot have persistent successors because the associated channel flux would tend to a positive constant. □

Theorem 5 (sharp minimum dimension). Under (T), (D), and strictly positive initial fractions, every vanishing state in a model with at most three states admits an exponential upper bound. The four-state example in Theorem 1 is therefore minimal.

Here the number of states counts population compartments in the channel model; it is not the dimension after eliminating conservation laws.

Proof. For three states, take a topological order (x, y, z). The final state z is a transition sink and persists. If x vanishes, Lemma 4 supplies a persistent dependency successor. Since x has no inflow, $\dot{x} \leq -cx$ eventually. If y also vanishes, its only possible persistent successor is z, and its inflow is bounded by Mx. Hence $\dot{y} \leq Mx - c'y$ eventually, for $c, c' > 0$. Variation of constants gives an exponential upper bound; any resonant factor t is absorbed by a slightly smaller exponent.

It remains to treat $x \to x^* > 0$ and $y \to 0$. A persistent-to-persistent channel flux cannot have a positive limit. Since x has no inflow, its outflow must therefore be αxy. Its contribution to the inflow of y is βxy, with $0 \leq \beta \leq \alpha$. Write the outflow from y as $y(\gamma x + \epsilon y + \delta z)$, with nonnegative coefficients. After using $z = 1 - x - y$, the exact two-coordinate equations are

$$\dot{x} = -\alpha xy, \qquad \dot{y} = y[dx - \delta + ey], \quad d = \beta - \gamma + \delta, \quad e = \delta - \epsilon. \tag{20}$$

If $\alpha = 0$, then $\beta = 0$, so y has no inflow and its persistent successor gives exponential decay. Suppose $\alpha > 0$. The limiting per-capita growth rate is $\ell = dx^* - \delta$. A positive ℓ contradicts $y \to 0$, and a negative ℓ gives exponential decay directly. We exclude $\ell = 0$.

Under that equality $d = \delta/x^* \geq 0$. Since x strictly decreases to $x^* > 0$, write the positive orbit as $y = Y(x)$ for $x > x^*$. Dividing (20) and using $Y(x^*) = 0$ gives

$$Y'(x) + \frac{e}{\alpha x} Y(x) = -\frac{d}{\alpha x}(x - x^*).$$

The coefficients are regular at x^*. Multiplication by $x^{e/\alpha}$ and integration yield

$$Y(x) = -\frac{d}{\alpha} x^{-e/\alpha} \int_{x^*}^{x} s^{e/\alpha - 1}(s - x^*)\, ds \leq 0, \tag{21}$$

contradicting positivity, including when $d = 0$. Thus $\ell < 0$.

With two states a vanishing state is a source, and the source-state argument applies. With one state no state vanishes. □

5. The complete four-state extinction boundary

The algebraic trajectory lies on a precise critical surface. This limits any claim of robustness and supplies a direct sensitivity calculation.

For arbitrary positive initial fractions in (3), define

$$K = I_0 + X_0 - Y_0, \qquad B = 2Y_0 - X_0, \qquad \Delta = B^2 - 4Y_0 I_0. \tag{22}$$

The same internal time gives

$$X = X_0 u, \quad Y = Y_0 u^2, \quad I = F(u) = K - X_0 u + Y_0 u^2, \qquad \dot{u} = -uF(u). \tag{23}$$

Theorem 6 (phase classification). Exactly one of the following occurs.

1. If $B > 0$ and $\Delta > 0$, then I vanishes exponentially. With

$$q = \frac{X_0 + \sqrt{\Delta}}{2Y_0} \in (0,1), \qquad \lambda = q\sqrt{\Delta} > 0,$$

   one has $u - q \sim ce^{-\lambda t}$ and $I \sim c\sqrt{\Delta}e^{-\lambda t}$ for some $c > 0$.
2. If $B > 0$ and $\Delta = 0$, then $q = X_0/(2Y_0) \in (0,1)$, $K = Y_0 q^2 > 0$, and

$$I(t) \sim \frac{1}{Kt^2}, \qquad \int_0^\infty I(t)\,dt = -\log q.$$

3. In every other case, $K > 0$ and $I \to K$. For some $c > 0$, $X \sim X_0 ce^{-Kt}$ and $Y \sim Y_0 c^2 e^{-2Kt}$.

In cases 1 and 2 the limit is $(X_0 q, Y_0 q^2, 0, 1 - X_0 q - Y_0 q^2)$; in case 3 it is $(0, 0, K, 1 - K)$. Both (T) and (D) hold in all three cases.

Proof. The quadratic F is positive at one and has positive leading coefficient. A first root in $(0,1)$, approached as u decreases, exists exactly when its vertex lies below one and its discriminant is nonnegative. These are $B > 0$ and $\Delta \geq 0$. For $\Delta > 0$, the first root is the larger root q and $F'(q) = \sqrt{\Delta}$. Separation of (23) gives the positive exponential equivalent in case 1. For $\Delta = 0$, $F(u) = Y_0(u - q)^2$; differentiating $1/(u - q)$ proves case 2.

If no root occurs in $(0,1)$, then $u \to 0$. Necessarily $K > 0$: a negative K forces an intervening root, while $K = 0$ makes $F(u) = u(Y_0 u - X_0)$ negative for small positive u. Now $\dot{u} = -Ku + O(u^2)$ gives $u \sim ce^{-Kt}$. The limits follow from (23) and conservation. In extinction cases the cycle $Y \leftrightarrow I$ has persistent successor Y from I; in the persistence case it has persistent successor I from Y. □

For $X_0 = Y_0 = 2/9$, the exact threshold is $I_0 = 1/18$. Values $I_0 = 1/36, 1/18, 1/12$, with R_0 adjusted to keep total mass one, lie respectively below, on and above the threshold. The graph is identical in all three cases. The critical equality is one equation in the initial fractions, so arbitrary perturbations need not preserve the algebraic tail. Multiplying every rate by a common positive constant changes the time unit and preserves the exponent.

6. An exact spectral repair

The graph conditions do not account for cancellation between limiting inflow and outflow. A transverse Jacobian does. The following statement uses (T) alone.

For a trajectory with limit y^*, let $V = \{i : y_i^* = 0\}$ be nonempty and P its complement. Put $v = y_V$, $p = y_P$, and

$$A = \left(\frac{\partial F_i}{\partial y_j}(y^*)\right)_{i,j\in V}. \tag{24}$$

A matrix is **Metzler** when its off-diagonal entries are nonnegative and **Hurwitz** when all its eigenvalues have strictly negative real parts. Write $s(A) = \max\{\Re\lambda : \lambda \in \sigma(A)\}$.

Theorem 7 (spectral equivalence). Under (T) and strictly positive initial fractions, A is Metzler and $s(A) \le 0$. Every vanishing coordinate admits an exponential upper bound if and only if $s(A) < 0$. In that case the entire trajectory converges exponentially to y^*.

Proof. Lemma 4's first part uses only (T). Each channel flux is integrable and tends to its value at y^*, so that limiting value is zero. Consequently every nonzero channel contains at least one factor indexed by V. Thus, for a constant C, $\|p'(t)\|_1 \le C\|v(t)\|_1$. The negative terms in F_i all contain y_i, so the transverse matrix has nonnegative off-diagonal entries.

Separate terms linear in v, whose other factor belongs to p, from quadratic terms in v. Since every $p_k^* > 0$, there are a positive diagonal matrix D and a nonnegative function $\eta(t) \to 0$ such that, componentwise for large t,

$$v' \ge (A - \eta(t)A_{\text{off}} - \eta(t)D)v. \tag{25}$$

Here A_{off} has the off-diagonal entries of A and zero diagonal. Indeed, choose η to dominate $\max_k |p_k/p_k^* - 1|$ and $\|v\|_1$. Each positive off-diagonal linear coefficient is at least $(1-\eta)$ times its limit. Changes in diagonal linear terms and all negative quadratic terms are bounded by a fixed diagonal multiple of η; positive quadratic terms may be discarded. Increasing D proves (25). If v decays exponentially, integration of the bound on p' also gives exponential decay of $p - p^*$, so η can be chosen integrable.

Suppose $s(A) > 0$. For sufficiently small fixed $\xi > 0$, the Metzler matrix $A_\xi = A - \xi A_{\text{off}} - \xi D$ has positive spectral bound. Eventually $v' \ge A_\xi v$. A nonnegative nonzero left Perron vector w of A_ξ gives $(w^T v)' \ge s(A_\xi) w^T v$. Strict positivity of v implies $w^T v > 0$, contradicting $v \to 0$. Thus $s(A) \le 0$.

If $s(A) < 0$, choose a strictly positive row vector $w^T = \mathbf{1}^T(-A)^{-1}$, so $w^T A = -\mathbf{1}^T$. The inverse is nonnegative, for example from $(-A)^{-1} = \int_0^\infty e^{At}\,dt$, and its diagonal entries are positive. Polynomial expansion yields $v' = Av + r(t)$ with $\|r(t)\|_1 = o(\|v(t)\|_1)$. Consequently $(w^T v)' \le -cw^T v$ eventually for some $c > 0$. This gives exponential bounds for v, and then for $p - p^*$.

Finally suppose $s(A) = 0$ and every vanishing coordinate decays exponentially. Take a nonnegative nonzero left Perron vector w with $w^T A = 0$ and put $q = w^T v > 0$. Since $w^T A_{\text{off}} = -w^T \operatorname{diag}(A)$, equation (25) gives

$$q' \ge -\eta(t) \sum_i w_i(D_{ii} - A_{ii})v_i \ge -L\eta(t)q \tag{26}$$

for a fixed positive L. Integrability of η then gives a strictly positive lower bound on $q(t)$, contradicting $q \to 0$. This excludes $s(A) = 0$ under exponential extinction. □

Metzler transverse Jacobians and strict spectral sufficient conditions are established tools in positive systems and boundary-stability theory [6-9]. Wiersdalen et al. [6, Theorem 2] give an attraction-versus-exponential-stability equivalence for a compartmental base matrix and compartmental perturbations preserving its zero off-diagonal and column-sum pattern. Johnston and Avram [7, Theorem 1] obtain local boundary-stability tests from strict reproduction-number inequalities. Adenane, Avram and Halanay [8, Theorems 6-7] study spectral thresholds and local

boundary bifurcations, with a Hurwitz tangential block and nondegenerate crossing conditions for the bifurcation result. Related balanced bilinear epidemic models are treated in [9].

The specific additional assertion here is the necessity of a strict transverse spectral gap for exponential approach along an initially positive trajectory in the acyclic channel class, including reducible transverse blocks. Every channel vanishes on the limiting persistent face, whose tangential dynamics are identically zero; the critical constructions therefore do not satisfy the tangential stability hypothesis of [8]. Our transverse matrix need not be compartmental, and the proof imposes no zero-pattern restriction on an additive perturbation. The criterion depends on the particular limiting state and supplies no graph-only procedure for determining it. We do not claim novelty for Metzler structure, positive-matrix comparison or the sufficiency of a Hurwitz block.

Corollary 8 (exact logarithmic extinction rate). Under the hypotheses and notation of Theorem 7,

$$\lim_{t\to\infty} \frac{\log \|v(t)\|_1}{t} = s(A). \tag{27}$$

Thus a zero spectral bound means that the total vanishing population has zero exponential rate, not merely that one chosen exponential estimate fails. If $s(A) < 0$, the same limit holds with $\|y(t) - y^*\|_1$ in place of $\|v(t)\|_1$. These are logarithmic rates; they do not assert a pure exponential equivalent or a common rate for each coordinate.

Proof. Fix $0 < \varepsilon < 1$. The lower comparison in (25) eventually gives

$$v' \geq A_\varepsilon v, \qquad A_\varepsilon = A - \varepsilon A_{\text{off}} - \varepsilon D.$$

A nonnegative nonzero left Perron vector w_ε satisfies $w_\varepsilon^T A_\varepsilon = s(A_\varepsilon) w_\varepsilon^T$. Since v is strictly positive, integration of the resulting scalar inequality gives $\liminf_{t\to\infty} t^{-1} \log \|v(t)\|_1 \geq s(A_\varepsilon)$. For the upper bound, the remainder estimate in Theorem 7 gives $|r_i(t)| \leq \|r(t)\|_1 \leq \varepsilon \|v(t)\|_1$ eventually for every i. Hence

$$v' \leq B_\varepsilon v, \qquad B_\varepsilon = A + \varepsilon \mathbf{1}\mathbf{1}^T.$$

The Metzler matrix B_ε has strictly positive off-diagonal entries (and the scalar case is immediate), so its left Perron vector can be chosen strictly positive. Its pairing with v is therefore comparable to $\|v\|_1$, and the upper scalar inequality yields $\limsup_{t\to\infty} t^{-1} \log \|v(t)\|_1 \leq s(B_\varepsilon)$. Continuity of the spectral bound as $\varepsilon \downarrow 0$ proves (27), including for reducible A.

If $s(A) < 0$, for every $\delta > 0$ with $s(A) + \delta < 0$ the upper comparison gives $\|v(t)\|_1 \leq C_\delta e^{(s(A)+\delta)t}$ eventually. Integrating $\|p'\|_1 \leq C\|v\|_1$ from t to infinity gives the same upper rate for $\|p - p^*\|_1$. The lower rate follows from $\|y - y^*\|_1 \geq \|v\|_1$. Letting $\delta \downarrow 0$ completes the proof. □

This corollary is a direct sharpening of the same positive-matrix comparison, not a claim to a new general theory of asymptotically linear systems. It is useful because it gives a precise aggregate rate even when the transverse block is reducible. Strict initial positivity remains essential: an identically absent coordinate can fail to excite the slow spectral mode.

In the four-state model, $A = [-\lambda]$ in the exponential-extinction regime, $A = [0]$ on the critical boundary, and $A = \operatorname{diag}(-K, -2K)$ when I persists and X, Y vanish. The aggregate logarithmic rates are respectively $-\lambda$, 0, and $-K$. In the family of Theorem 2 only I vanishes; identity (17) gives its transverse eigenvalue zero.

7. Verification, interpretation and limits

The executable `verify.py` uses only Python's standard library and exact rational arithmetic. It constructs each ordinary differential equation from the individual conversion channels, independently constructs the closed-form polynomial trajectory, and checks equality of every coefficient after differentiation. It also checks normalisation, exact limits, transition acyclicity and condition (D). The latter is checked by deleting all vertices with a persistent dependency successor and testing the remaining induced graph for acyclicity. All 24 cases $m = 2, \dots, 25$ pass. The three phase regimes are separately checked against the channel equations. These finite checks detect algebraic and transcription errors; they do not replace the proofs for arbitrary m or the minimum-dimension theorem. The separate plotting script evaluates the scalar quadrature using NumPy, SciPy and Matplotlib; these packages are not required by the exact verifier.

The scalar time change explains the mechanism. For a positive activity I, the accumulated activity $\tau = \int I$ makes the reservoir equations linear. In Theorem 2, $u = e^{-\tau}$ and $I = C(u-q)^m$ for $q = 1/2$. Physical time diverges when the internal-time activity reaches its repeated zero. More generally, if an analytic scalar reduction satisfies

$$\tau' = F(\tau), \qquad F(\tau) = a(\tau^* - \tau)^m + o((\tau^* - \tau)^m), \quad a > 0,$$

at a finite terminal internal time, a simple zero gives exponential decay; an m-fold zero with $m \geq 2$ gives an activity proportional to $t^{-m/(m-1)}$. This is a scalar asymptotic calculation. The construction shows that the required cancellations can occur inside the original nonnegative quadratic channel class.

The result provides no fitted parameters, observed-population evidence or recommendation for disease control. Its applied relevance is mathematical: a modeller cannot use the two graph conditions alone to justify an exponential tail, an exponential stopping rule or a logarithmic extinction-time estimate. Under the exact family, reaching a small activity threshold a instead takes asymptotically $(K_m/a)^{1/p_m}$ time, whereas an exponential law would give logarithmic growth in $1/a$. Any practical use requires matching the model, its parameters and its initial condition.

The four-state slow tail requires a critical equality. Higher multiplicities impose further cancellations. We have proved existence and exact consequences, not prevalence or robustness to arbitrary perturbations. Theorem 7 supplies a strict spectral condition under which exponential convergence does follow. The bounded search through 19 October 2026 found no earlier resolution of the source conjecture, but cannot certify historical priority or exclude unindexed work. The proof does not depend on such a priority claim.

8. Availability and declarations

Data and code availability. This is a theoretical study with no observational dataset. The accompanying repository contains this manuscript, `verify.py`, `test_verifier.py`, exact replay receipts, the figure and its plotting source. The source repository is https://github.com/ipitchford/algebraic-extinction-compartments and the versioned archive is https://doi.org/10.5281/zenodo.23501636. Running `python3 -B test_verifier.py --output-dir NEW_DIRECTORY` repeats the finite rational checks in normal and optimized Python modes and requires three deliberate corruptions to be rejected in each mode. The analytic proofs, rather than those finite checks, establish the general results.

Ethics. No human participants, animal subjects or personal data are involved. The models are mathematical constructions, not clinical or policy recommendations.

Contributions. The release records conceptualisation, formal analysis, software, validation, visualisation and writing. It is presented anonymously; the supplied research record does not assign these CRediT roles to named human contributors.

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AI assistance and assurance. AI assistants were used for research development, proof criticism, code and manuscript preparation. This revision used Codex for proof inspection, exposition, a logarithmic-rate corollary and verification hardening, with a separate AI check of the corollary. AI review is not external human peer review. Exact arithmetic replay is not proof-assistant formal verification, and the bounded literature search does not certify historical priority.

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